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Machine Learning for Business Applications

Week 2 : the big picture

How LLM labeling works in business, and what we use in this course

The big picture

A typical labeling pipeline

1. **Collect**
   Comments, tickets and reviews arrive from a website, an app or email
2. **Write the guide**
   Categories, definitions and rules for hard cases. This becomes the prompt
3. **Label**
   A model reads each comment and returns a category
4. **Check**
   Validate the answer, compare with people, flag what looks wrong
5. **Use**
   Charts, alerts, routing to a team, reports for managers

The model is one step out of five.

The big picture

Who does the labeling in real companies?

OPTION	WHAT IT IS	TRADE-OFF
People	Staff or an outside vendor read and label	Accurate, but slow and costly
Rules	For example: contains the word "refund"	Fast, but only catches what you thought of
A trained model	A classifier built from many labeled examples	Needs lots of labeled data first
An LLM through a paid API	A hosted model you call over the internet	Flexible, but costs per use; data leaves the company
An LLM the company runs	An open model on the company's own servers	Data stays in-house, but someone must run it

Real systems often mix them: people check a sample.

The big picture

What a company weighs

FACTOR	THE QUESTION
Cost	What does it cost for 10,000 comments a day?
Privacy	Is the customer text allowed to leave the company?
Quality	What does a wrong label cost?
Speed	Do we need the answer now, or tonight?
Effort	Who builds it, and who keeps it running?

There is no single right answer. It depends on the business.

What we do in this course

IN A COMPANY

Whatever fits the budget

A paid API or a vendor
Security and privacy review
Monitoring and support

IN THIS COURSE

Free options

A free LLM API: Groq
A free model: openai/gpt-oss-20b
Python in Colab, apps in Streamlit

The steps are the same. Only the budget is different.

This course

Our pipeline, next to a real one

STEP	IN A COMPANY	IN THIS COURSE
Collect	Website, app, support system	**A comment typed into our app**
Guide	A prompt written by the team	**A prompt you write**
Label	A paid API or the company's own model	**Groq free API, openai/gpt-oss-20b**
Check	Human review, monitoring	**A check on the answer, and a flag when the label and rating disagree**
Use	Dashboards, alerts, routing	**Charts and a "read these carefully" table**

A small app, but the same five steps.

This course

Swapping the model is a small change

```
URL   = "https://api.groq.com/openai/v1/chat/completions"
KEY   = (your secret key)
MODEL = "openai/gpt-oss-20b"
```

Another provider usually means changing these three lines. Your guide and the rest of the code stay the same.

Check the provider's documentation: the details differ between companies.

The guide is yours. The model is a part you can swap.

This course

What free means

GOOD FOR

Learning and prototypes

Trying ideas at no cost

Small datasets

No contract, no card

NOT FOR

A real business system

Rate limits and no guarantees

Models get retired and renamed

No promise about your data

If the model name stops working, check the provider's model list.